

Lecture 16: The Physics of Control (Delayed Neutrons and the Inhour Equation)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Section 7.2** Reactor Kinetics (Prompt/Delayed Neutrons, Inhour Equation, Prompt Critical).

Recap: The Problem with Prompt Neutrons

In Lecture 15, we saw that if a reactor relied solely on prompt neutrons, the time scale of the reaction (the Reactor Period T) would be on the order of milliseconds.

$$T \approx \frac{l}{\rho} \quad \text{where} \quad \rho = \frac{k-1}{k}$$

If $l = 10^{-4}$ s and $\rho = 0.001$, the power rises by a factor of 22,000 in one second. This is uncontrollable. Today, we mathematically prove how **Delayed Neutrons** save us.

1 The Point Kinetics Equations

To account for delayed neutrons, we must modify our neutron balance equation. We split the neutron source into two terms:

1. **Prompt Neutrons:** $(1 - \beta)$ fraction of fission neutrons appear instantly.
2. **Delayed Neutrons:** β fraction of fission neutrons appear later, coming from the decay of Precursors (C).

We now need two coupled differential equations: one for the Neutrons (n) and one for the Precursors (C).

1.1 1. The Neutron Equation

$$\frac{dn}{dt} = \underbrace{\frac{k(1-\beta)n}{l}}_{\text{Prompt Generation}} + \underbrace{\lambda C}_{\text{Delayed Source}} - \underbrace{\frac{n}{l}}_{\text{Loss}}$$

Here, λ is the decay constant of the precursor (e.g., ^{87}Br).

1.2 2. The Precursor Equation

The precursors are created by fission and lost by radioactive decay.

$$\frac{dC}{dt} = \underbrace{\frac{k\beta n}{l}}_{\text{Production}} - \underbrace{\lambda C}_{\text{Decay}}$$

2 The Solution: The Inhour Equation

We look for solutions of the form $n(t) = n_0 e^{t/T}$ (exponential growth with period T). Substituting this ansatz into the coupled system involves some algebra (omitted here for brevity, see Lamarsh Ch. 7), but the result is the fundamental equation of reactor control:

$$\rho = \frac{l}{Tk_{eff}} + \frac{\beta_{eff}}{1 + \lambda T} \quad (1)$$

The Inhour Equation (Simplified One-Group Form)

Historical Note: You might wonder who "Dr. Inhour" was. There is no such person! The term "Inhour" is actually a portmanteau of **In**verse **H**our. It was an early unit of reactivity defined by Enrico Fermi's team at Chicago Pile-1 as the amount of positive reactivity that would result in a stable reactor period of exactly one hour.

This equation relates the **Reactivity** (ρ , what the operator changes with control rods) to the **Reactor Period** (T , how fast the power changes). Recall the reactor period T is the time required for power to change by a factor of e .

Note: In a real reactor, there are 6 distinct groups of precursors with different decay constants λ_i . The full equation is a sum:

$$\rho = \frac{l}{T} + \sum_{i=1}^6 \frac{\beta_i}{1 + \lambda_i T}$$

3 Analyzing the Result

Let's analyze the One-Group equation to understand the three distinct regimes of reactor operation.

3.1 Regime 1: Sub-Prompt Critical (Normal Operation)

Suppose we insert a very small amount of reactivity: $\rho \ll \beta$. Assume $\rho = 0.001$ (0.1%). For U-235, $\beta \approx 0.0065$ (0.65%). Since l is tiny (10^{-4}), the first term l/T is negligible for large T . The equation simplifies to:

$$\rho \approx \frac{\beta}{1 + \lambda T}$$

Solving for Period T :

$$T \approx \frac{\beta - \rho}{\lambda \rho}$$

Using $\lambda \approx 0.1 \text{ s}^{-1}$ (average precursor decay constant):

$$T \approx \frac{0.0065 - 0.001}{(0.1)(0.001)} = \frac{0.0055}{0.0001} = \mathbf{55 \text{ seconds}}$$

Result: Instead of 0.1 seconds (Prompt), the period is 55 seconds. This is slow! A human operator can easily watch a gauge and adjust a rod in 55 seconds.

3.2 Regime 2: The Threshold (Prompt Critical)

What happens if we add enough reactivity such that $\rho = \beta$?

$$\beta \approx \frac{l}{T} + \frac{\beta}{1 + \lambda T}$$

The math blows up. T must become very small for this equality to hold. Physically, this means the reactor is critical on prompt neutrons *alone*. The delayed neutrons are no longer needed to sustain the chain reaction.

3.3 Regime 3: Super-Prompt Critical (The Danger Zone)

If $\rho > \beta$, the reactor is supercritical on prompt neutrons.

$$\rho \approx \frac{l}{T}$$

We return to the "Bomb" kinetics of Lecture 15. The period drops to milliseconds with only a tiny increase in the reactivity.

$$T \approx \frac{10^{-4}}{0.001} = 0.1 \text{ s}$$

3.4 Regime 4: Large Negative Reactivity (The SCRAM)

We have discussed what happens when you pull rods out, but what happens when you slam them all in to shut down the reactor (a SCRAM)?

Suppose we drop the control rods, inserting a massive negative reactivity of $\rho = -\$10.00$.

Looking at the physics of the "Storage Buffer":

1. **The Prompt Drop:** The prompt neutron population, which has a lifetime of 10^{-4} seconds, collapses instantly. The reactor power rapidly drops by a few decades in a fraction of a second.
2. **The Precursor Tail:** The power does not drop to zero! The precursors that were created before the SCRAM are still sitting in the fuel. They continue to decay, releasing delayed neutrons into the subcritical core. A nice plot of the power after SCRAM can be found at www.nuclear-power.com.

If we look at the extreme limit of the Inhour equation for large negative reactivity, the period T approaches the mean life of the longest-lived precursor group. The longest-lived precursor is Bromine-87, with a half-life of 55 seconds (which corresponds to a mean life $\tau = 55/\ln(2) \approx 80$ seconds).

Result: No matter how much negative reactivity you insert, once the prompt drop is over, the reactor power will decay on an unchangeable **-80 second period**. The reactor is chained to the physics of Bromine-87. Once the prompt population is gone, the remaining power decay rate is fundamentally limited by delayed-neutron precursor decay.

4 Effective Delayed Neutron Fraction (β_{eff})

In the Inhour equation, you might have noticed the term β_{eff} instead of just β . Why is the "effective" fraction different from the physical fraction of delayed neutrons?

It comes down to the energy of the neutrons at birth:

- **Prompt Neutrons** are born at an average energy of ~ 2.0 MeV.
- **Delayed Neutrons** are born from the decay of precursors at a much lower average energy of ~ 0.4 MeV.

Because delayed neutrons are born at lower energies, they do not have to migrate as far through the moderator to reach thermal energies. Using the leakage concepts from last week, delayed neutrons have a smaller Fermi Age (τ_T), which means they have a significantly higher **Fast Non-Leakage Probability** (P_{FNL}).

Because they are less likely to leak out of the core, a single delayed neutron is actually "worth more" to the chain reaction than a single prompt neutron! We define an Importance Factor (γ):

$$\beta_{eff} = \gamma\beta \quad (2)$$

In a large commercial Light Water Reactor, $\gamma \approx 1.04$. However, in a small, highly leaky research reactor, γ can be as high as 1.20, meaning the delayed neutrons are 20% more effective at sustaining the chain reaction than the prompt neutrons.

5 Units of Reactivity: The Dollar

Because the fraction β is the most important safety limit in a reactor, we use it as a unit of measurement.

- **1 Dollar (\$1.00)** = A reactivity of β .
- **1 Cent (1¢)** = $0.01 \times \beta$.

For U-235 ($\beta = 0.0065$):

- \$1.00 = $0.0065\Delta k/k$
- \$0.50 = $0.00325\Delta k/k$ (Safe)
- \$1.10 = $0.00715\Delta k/k$ (Prompt Critical - Dangerous)

"The reactor is rising on a 50-cent period" is standard control room terminology.

5.1 Case Study: The Plutonium Problem

To understand why we use Dollars instead of just reporting absolute $\Delta k/k$, imagine two identical operators working on two different reactors. Operator A is running a standard U-235 reactor. Operator B is running an experimental Pu-239 Fast Reactor.

Both operators make the exact same minor error: they accidentally withdraw a rod that inserts an absolute reactivity of $\Delta k/k = 0.004$ (just 0.4%).

- **Operator A (U-235):** Since $\beta = 0.0065$, the reactivity is $\rho = 0.004/0.0065 = \$0.61$. Looking at the Inhour equation, this puts the reactor on a stable, easily manageable period of about 10 seconds. Operator A calmly inserts the rod back in.

- **Operator B (Pu-239):** Since $\beta = 0.0021$, the exact same rod movement yields $\rho = 0.004/0.0021 = \$1.90$. Operator B just went prompt critical, and the core melts in milliseconds.

The Lesson: Absolute reactivity ($\Delta k/k$) doesn't tell you how dangerous a reactor state is unless you know the fuel's specific β . 50 cents is identically safe in both reactors, even though the physical rod movements required to get there are vastly different.

6 Physical Insight: Why does it work?

Think of the delayed neutrons as a "storage buffer."

- When you increase reactivity, the prompt neutrons rise instantly.
- But the delayed neutrons (0.65% of the population) are "stuck" in the precursors (Bromine/Iodine atoms). They haven't been born yet.
- The total population cannot rise to the new equilibrium level until those precursors decay and release their neutrons.
- This "drag" holds the reactor back, forcing it to wait for the precursors (approx. 10 to 80 seconds).

7 Summary

1. **Point Kinetics:** We couple the neutron population to the precursor concentration.
2. **The Inhour Equation:** Relates Reactivity (ρ) to Period (T).
3. **Safety Limit:** We must keep $\rho < \beta$ (less than \$1.00).
4. **Normal Operation:** As long as $\rho < \beta$, the reactor period is determined by the slow decay of precursors (λ), making the reactor controllable by humans.
5. **General Reference:** A very accessible reference for reactor kinetics is the DOE Handbook [Nuclear Physics and Reactor Theory](#). I've found it a bit easier to read than Lamarsh (which it draws heavily from).